

Introduction to Contract Theory and Auctions

Introduction

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- Arrow-Debreu model does not apply if a trading partner has private information about some *common value* (Akerlof 1970)
- E.g., Buyer draws inferences about the quality of a car from the mere fact that the seller supplies it.
- Details of the market environment matter, and typically game theory has been used to predict market outcomes.
- This revolution has deeply changed our views on how insurance, financial and labor markets work (Riley 2001).

- Exchange of goods in perfectly competitive frictionless markets
 - Perfect information
 - No externalities
 - No strategic interaction
- Rational agents take market prices as given to decide on actions
- Market prices adjust to equilibrate demand with supply
- Its theoretical apex: general equilibrium theory
 - Existence
 - Efficiency
 - First welfare theorem: Equilibrium is efficient
 - Second welfare theorem: Any efficient allocation can be achieved as an equilibrium after appropriate lump-sum transfers
- Later reformulated to account for uncertainty but in a rather uninteresting way
 - von Neumann and Morgenstern (1944): Preferences over uncertain outcomes
 - Arrow (1964) and Debreu (1959): State contingent commodities

- Why is there unemployment?
- Why does equity play such a small role in financing new investment?
- Why do firms pay dividends even though they are double-taxed?
- Why do firms offer lower premiums to insurees who accept higher deductibles?
- Why does an announcement of stock repurchase cause stock price to rise?
- many many more...

- Standard paradigm: Imperfect information is only a transaction cost
- Information economics: No!
- Small imperfections in information can have large consequences
- Why?
 - Agents' choices reveal information
 - When they choose an action they consider not only what is good for them but also how it changes others' beliefs about them
- Standard paradigm: large number of agents (competitive)
- But, as soon as a small number of agents engage in a relationship and they make relationship-specific investments, they enter into a small-number bargaining situation. Does this lead to inefficiency?
- No according to Coase: *“Coase Theorem” in the absence of “transaction costs” the outcome of private bargaining is efficient.*

- During the last 50 years or so most exciting developments in economics happened in information economics.
- Information economics analyzes
 - Consequences of informational asymmetries between actors
 - Optimal design of contracts, institutions, etc. to deal with them
- Examples:
 - Insurance companies have imperfect information about the risk characteristics of their clients
 - Adverse selection
 - Screening
 - Firms do not know the abilities of job applicants
 - Signaling
 - Firms cannot observe the effort of their workers
 - Moral Hazard

A simple example

- You are interested in taking over an existing company
- The value of the company for the current owner: V dollars
- It is 50% more valuable for you: $1.5V$
 - You are willing to pay any price up to the expected value
- You have all the bargaining power
 - If your offer is at least as large as the expected value for the current owner, it will be accepted
- Three possible scenarios:
 - ① Perfect information
 - ② Imperfect but symmetric information
 - ③ Asymmetric information

A simple example

- ① Perfect information: Both know V
- ② Imperfect but symmetric information: Both believe V is uniformly distributed between 0 and 100
- ③ Asymmetric information: Current owner know V , you believe it is uniformly distributed between 0 and 100
 - Inefficient outcome
 - Acceptance of an offer reveals information
 - This influences your preferences
 - Illustrates what is known as **adverse selection**
 - Offer (adversely) selects only those with low value

- Health insurance
 - Suppose insurance company offers a package tailored for average risk in the population
 - Only riskier people buy → average risk in the insurer pool higher than that in general population
 - Company may lose money
 - Known as **adverse selection**
- Can the company do something about it?
- Offer different packages and let people choose
 - High premium + Low deductible: High risk
 - Low premium + High deductible: Low risk
- Known as screening
- Other examples of this sort:
 - Price discrimination by a monopolist
 - Regulation of monopolies
 - Financial contracts and credit rationing

- Risk averse entrepreneur seeking finance for a risky project
- Entrepreneur has better information than lender about the risk-return profile of the project
- May cause many worthy projects not to be financed (credit rationing)
- Can an entrepreneur with a good project do something about it?
- Can put her money where her mouth is
 - Can retain some equity
 - This is costly for a risk averse entrepreneur
 - But less so for one with a better project
- This action signals private information
- Such models are known as **signaling** models
- Other examples include:
 - Education as a signal of ability
 - Warranty signals quality

- More insurance coverage → less effort to prevent accidents
- This is known as moral hazard in insurance (and now in economics)
- If you receive a fixed grade for this course you may not work as hard
- Models that include such concerns are also known as **principal-agent** or **agency** models
- General feature:
 - agent takes an action that affects both the principal and the agent
 - only an imperfect indicator of agent's action is observed
- What is the optimal (compensation) contract?
- Two factors to take into account:
 - ① Incentives: Payment must be sensitive to outcome to induce effort
 - ② Risk sharing: If agent is more risk averse, he should not be exposed to too much risk

- In standard competitive model contracts are very simple:
 - I will pay a certain amount if you do such and such
- In real world they are much richer
- Compensation contracts
 - depend on absolute as well as relative performance
 - may also include fixed payments plus stock options
- Financial contracts
 - specify not only interest rates but also collateral requirements
 - control rights
- Economics of information help us understand why
 - hence known also as **Contract Theory**

An impressive number of diverse areas including

- Design of institutions and organizations
- Corporate finance
- Regulation, auditing
- Development economics
- Macroeconomics
- Public finance: optimal taxation
- Allocation of ownership: vertical integration, privatization
- Accounting
- Law
- Political Science

Why do we care about asymmetric information?

- Both moral hazard and adverse selection have been recognized as severe problems by the insurance industry
- We see their workings everyday
- Savings and Loan crisis in the US
 - Government insured savings deposits of up to \$100,000
 - Newly deregulated S&Ls made riskier investments than they would have otherwise
 - During the 80s and 90s, 747 S&Ls failed
 - Cost: \$160 billion, of which government paid \$125 billion
- Similar issues during 2007 financial crisis
 - Financial institutions and investors have asymmetric information about the riskiness of financial securities
 - The US government bailed out Fannie Mae and Freddie Mac and AIG
 - Should it have bailed out Lehman Brothers?
 - Moral hazard is cited as an important consideration in this decision

- To a Large extent synonymous, meaning “Rules of the game”.
- Describe what actions can parties can undertake, and what outcomes these actions would entail.
- Almost everything in everyday life is defined by institutions: legal or social.
- Without rules or institutions we would be in the “natural state”
- Fortunately, the rules of the game are written by someone in most cases. Usually to achieve better outcomes.
- Mechanism design= normative economics.
- Knowledge of Game Theory is important because it predicts how a game will be played by agents.
- Mechanism design takes the physical environment as given and the constraints faced by the designer (incentive) and asks what mechanisms are feasible or optimal?
- Mechanism design: how can we improve upon a particular institution or design a new one?

- Contract Theory vs Mechanism Design: MD is mathematically elegant and abstract, but has been unable to answer many important questions (socialism vs capitalism for example).
- MD has proven useful for tackling smaller questions, specifically, business practices, i.e. contracts.
- In this reincarnation as **contract theory** we have that
 - The contract is designed by one of the parties.
 - Parties may refuse to participate.
 - Some theoretical shortcuts are taken, e.g. restricting attention to simple or incomplete contracts.
- However, recent work in mechanism design has provided a theoretical foundation for simple or incomplete contracts.

A Taxonomy of Models

		Source of Imperfect Information	
		Characteristic	Action
Who moves first	Uninformed	Screening and Adverse Selection	Moral hazard
	Informed	Signaling	?